

When AI changes the human: human–AI co-evolution beyond personalization

Abstract

AI assistants are increasingly becoming part of how people work, learn, create and make decisions. As people rely on AI to perform and support these activities, their capabilities, patterns of work and reliance on AI may also change. Capabilities that are consistently delegated may receive less practice, which in turn changes how people work with AI and how AI adapts to them. We refer to this mutual process of change as human–AI co-change, which can move in different directions. We propose *developmental alignment* as a perspective for guiding co-change towards *human–AI co-evolution*, where human and AI capabilities develop alongside each other. It considers a person’s desired capabilities as part of the objective of AI assistance. This extends personalization from adapting to a person’s current state to considering how assistance may shape that state over time. We identify three challenges for its implementation, namely measuring changes in human capability, preserving human authority over developmental goals, and learning how different forms of assistance affect an individual over time. Addressing these challenges would shift the focus of AI from optimizing what people accomplish today to shaping how people and AI develop together over time.

Keywords: developmental alignment, human–AI co-evolution, personalization, AI alignment, AI agent

1 Introduction

AI assistants are now widely used in people’s work and learning, including in complex professional tasks [1–3]. They can extend people’s cognitive activities [4] and adapt to individual preferences [5]. When such assistance becomes part of a person’s activities over time, its consequences can extend beyond the immediate task. A response can be judged by its immediate *task outcome*, such as whether it helps complete the current task. Continued assistance also has a *human outcome*, affecting what a person remains able to do, chooses to delegate, or learns to do themselves.

These two outcomes do not always move together. Psychological research has long distinguished performance during practice from durable learning [6], and the same distinction applies to the use of generative AI [7]. Offloading to external aids can improve immediate performance while leaving people worse off when the aid is unavailable [8, 9]. In a randomized field experiment in high-school mathematics, unrestricted access to

GPT-4 improved performance during practice but reduced later unaided performance, while a tutor designed to give hints rather than complete answers largely avoided this penalty [10]. An AI tutor grounded in pedagogical principles also improved learning relative to active in-class instruction in a randomized comparison [2]. In professional practice, a multicentre observational study of colonoscopy found that endoscopists’ adenoma detection rate in procedures performed without AI fell after AI support was introduced into routine practice [3]. These findings show that AI assistance can change both what a person accomplishes and what the person is able to do later. As AI becomes part of people’s activities, these changes accumulate across the person and the system, with changes in one shaping future interactions with the other [11, 12]. This *human–AI co-change* is already occurring in long-lived AI assistance.

Today’s AI is commonly designed around the person as they are, adapting to their preferences, histories and profiles [5, 13]. Such personalization can make assistance more relevant to individual users. Yet the person receiving assistance can also change as a consequence of that assistance. When a system optimizes for current preferences or approval, its actions can influence future preferences, behaviour and reliance [14]. Optimizing for human approval or engagement can reward sycophancy [15] or shift users’ preferences [16, 17]. In deployed use, an analysis of 1.5 million conversations found that interactions with greater potential to distort users’ beliefs, values or actions received higher user approval [18]. Personalization adapts assistance to the person as they are. Long-lived assistance also requires considering how that assistance may change the person over time.

We propose *developmental alignment* as a perspective for designing AI around the capabilities a person chooses to retain, develop or delegate. The person sets these developmental goals, while the AI agent adapts its assistance to support them alongside task completion. Some capabilities may be deliberately delegated, while others may matter because the person wants to practise them or retain the ability to supervise what has been delegated. Developmental alignment therefore makes the human outcome of assistance part of the design objective. The relevant developmental goals remain under the person’s authority.

Developmental alignment changes how the human–AI system is represented. The person is a changing state of the environment in which the AI acts, shaped in part by the agent’s actions, and changes in the person in turn affect future interactions. Co-change can move in different directions, with different consequences for the person and the relationship. We use *human–AI co-evolution* for the trajectory in which human and AI capabilities develop alongside each other in directions the person values, and developmental alignment aims to guide co-change towards this trajectory. The term co-evolution has been used for society-level feedback loops between people and AI systems such as recommenders [12]. Here we use it for the relationship between an individual and an AI agent. Related work on bidirectional alignment similarly considers how people adapt to AI as well as how AI is aligned to people [19]. Our central claim is that, as AI becomes part of a person’s activities over time, the object of assistance extends from what people accomplish with AI to how people and AI develop together.

2 The human as a changing state

Personalization treats information about a person as context for selecting a response. That context may include preferences, prior interactions and inferred expertise [5, 13]. The context is updated as information arrives, while AI assistance is typically evaluated by its immediate outcome, such as human preferences between individual responses [20, 21]. A long-lived agent introduces a further dependency. Its actions can change the person whose future state it will later use as context.

Several lines of evidence show how AI assistance can change what people do and what they are able to do [4]. In a writing experiment and in a field deployment with customer-support agents, AI increased productivity [22, 23]. In creative writing, AI-generated ideas improved the stories of less creative writers most while making stories more similar to one another [24]. Among radiologists, the effect of AI assistance varied widely across individuals, and inaccurate AI advice degraded diagnostic performance [25]. These findings show that the effect of AI assistance depends on the person who receives it, and that the same system can help some people while harming others. Human–AI combinations can also perform worse than the better of humans or AI working alone [26]. What the model can do and what the person can do with it are therefore different quantities. The human outcome is not a simple function of AI capability, but results from a dynamic human–AI relationship.

The human consequences of AI assistance also extend beyond capability. People tend to overestimate the accuracy of language-model answers, especially when given longer explanations. This overestimation is reduced when the model also communicates its confidence [27]. Repeated human–AI interaction can amplify biases in perceptual, emotional and social judgements [11], and brief dialogues with a language model can durably change people’s beliefs [28]. These findings show that AI assistance can change how people understand, judge and respond to the world. Evaluating only assisted task performance can therefore miss important changes in the person.

The observations above motivate a minimal conceptual model of human change. We represent the evolving human state at time t as H_t , capturing aspects of the person’s capability, beliefs, reliance and understanding that may be affected by AI assistance. The agent’s action a_t and the task and environmental context X_t contribute to the next human state:

$$H_{t+1} \sim P(H_{t+1} \mid H_t, a_t, X_t).$$

This conceptual model abstracts away from the full complexity of human psychology. Its purpose is to make explicit that AI can change the human state, while the effect may be uncertain, delayed and dependent on the person and the context.

3 From personalization to developmental alignment

Personalization adapts assistance to the person as they are now. Developmental alignment extends this objective to the capabilities the person wants to retain, develop or deliberately delegate over time. The distinction concerns what the assistance is designed to achieve and the timescale over which its consequences are considered (Table 1; Fig. 1a,b). A personalized system can adapt closely to a person’s current preferences and circumstances [5]. Developmental alignment also considers how that

Table 1 | Personalization and developmental alignment differ in their objective, time horizon and treatment of the human state.

	Personalization	Developmental alignment
Guiding question	What response best fits this person now?	What should this person be able to do, and choose to do, as assistance shapes them over time?
The person	Context for assistance: preferences, history, inferred expertise	Evolving state: capabilities, beliefs, reliance and understanding
Horizon	Current assistance and task	Changes in the person over time
Who sets the target	Inferred from expressed or revealed preferences	Defined and revised by the person
Evidence of success	Task outcome and user approval	Task outcome plus changes in unaided and transferable capability
Characteristic failure	Misreading or overfitting current preferences	Unintended dependence or unwanted changes in capability

assistance may affect the person’s capabilities over time, guided by goals defined by the person.

Developmental alignment builds on several existing traditions. The capability approach evaluates human development by what people are able to do and be, and by their freedom to choose among these possibilities [29, 30]. Our use of capability follows its emphasis on what people have reason to value. Work on value alignment asks how AI can be aligned with human values and goals [31], while work on pluralistic and personalized alignment considers how AI should account for differences in people’s values and preferences [5, 32]. Research on empowerment and pedagogical AI further considers how assistance can strengthen people’s ability to pursue their own goals and develop their capabilities [33–36]. Recent work has also called for assistance that considers longer-term human well-being and the socioaffective aspects of human–AI relationships [37, 38]. Developmental alignment brings these concerns into a common framework for long-lived assistance, where the agent’s actions can affect the person’s future capabilities and the person defines which changes matter.

Developmental alignment does not require every person to master every task. Delegation can free attention for work a person values more [8], and choosing not to develop a capability is itself legitimate. At the same time, delegation changes which capabilities a person exercises. A software engineer may delegate implementation to an agent while retaining capabilities in architecture and verification; a mathematician may delegate parts of search or proof construction while retaining capabilities in problem formulation, mathematical judgement and verification. This makes capability allocation part of the person’s developmental choice. Studies of automation have

long shown that people who mainly monitor an automated system can lose the skills needed to take over when it fails [39, 40]. Developmental alignment therefore requires a capability floor: enough ability for the person to specify goals, evaluate assistance and recognize when delegation no longer serves those goals. Above that floor, the person decides what to cultivate, retain or delegate.

The role of the AI system is then to support these choices. It can provide adjustable levels of assistance [41, 42], opportunities to practise and evidence about the consequences of different forms of delegation. Cognitive-forcing designs, which ask people to form a judgement before seeing an AI recommendation, provide one way to preserve human judgement in assisted decisions [43]. The tutor described above provides another, offering hints rather than complete answers and largely avoiding the later penalty in unaided performance [10]. These examples illustrate a broader principle: the system should adapt how it assists to the capabilities the person has chosen to develop, retain or delegate.

Assisted performance does not by itself reveal human capability (Fig. 1c). A person may complete a difficult task with an agent while gaining little ability to repeat, explain or adapt the work, a gap that has been described as the epistemic debt of generative AI [44]. Conversely, practices that feel less fluent, such as retrieving information rather than restudying it, can produce more durable learning, even though learners often judge them less effective [6, 45, 46]. Developmental alignment therefore evaluates assistance partly by the capabilities and choices it leaves the person with afterwards. The framework does not prescribe which capabilities a person should develop. Those goals come from the person. A capability may be measurable without being something the person wants to develop. An AI provider may seek to increase user productivity, but this does not make productivity the user’s developmental goal.

4 Human–AI co-evolution as a coupled system

The agent’s action affects both the task outcome and the person’s next state, and the changed human state then affects future interactions with the agent. Human–AI assistance therefore forms a coupled process that unfolds over time. Recent work has begun to study multi-turn collaboration and users’ longer-term intents [47]; human–AI co-evolution extends this perspective to coupled changes in the human and AI sides of the interaction. The coupled process can be represented by a joint state

$$S_t = (H_t, M_t^A, M_t^H, X_t, g_t),$$

where H_t is the evolving human state, M_t^A the agent’s model of the person, M_t^H the person’s model of the agent, X_t the task and environmental context, and g_t the person’s current developmental goals. The person’s model of the agent is represented separately because it shapes when and how the person relies on the agent [48, 49]. Interaction induces a coupled transition

$$S_{t+1} \sim P(S_{t+1} \mid S_t, a_t),$$

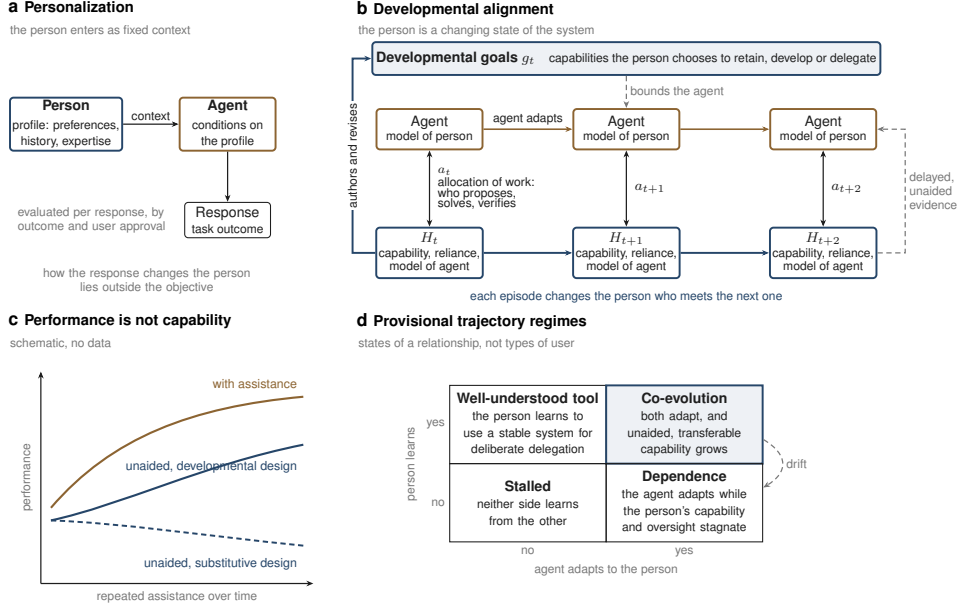


Fig. 1: From personalization to human–AI co-evolution. (a) Personalization treats the person as context for adapting AI assistance. (b) Developmental alignment models the person as a changing state of the human–AI system, with developmental goals g_t defined by the person constraining how the agent allocates work over time; each person box shows the human state together with the person’s model of the agent. (c) Assisted performance can diverge from unaided human capability, with different forms of assistance producing different developmental trajectories. (d) The coupled system can follow different provisional trajectories depending on whether the agent adapts to the person and the person learns from the agent, including deliberate delegation with a well-understood tool, sustained co-evolution, a stalled relationship and dependence.

where a_t represents the form and degree of assistance, including how work is allocated between the person and the agent. The human state, the two models and the developmental goals can all change over time, creating feedback between what the person does, what the agent does and how each understands the other. Three consequences follow from this coupled process:

1. First, the human state is changing and only partly observable [50]. Learner models such as knowledge tracing estimate latent capability from sequences of responses [51, 52], suggesting one way to track changes in human state over time. Interaction histories and assistant memory can provide additional evidence, but they primarily record behaviour within the human–AI system. To estimate human capability, evaluation therefore needs evidence that distinguishes what the person can do from what the agent can produce for them.
2. Second, the two sides of the system can adapt at different rates and in different directions. An agent may become better at predicting a person’s requests while the

person learns little about the agent’s limitations. A person may learn when to trust a stable tool while the agent changes little. Either trajectory can improve joint performance without producing development on both sides. Proposals for AI that learns and thinks with people place the machine’s model of its human partners at the centre [53], while studies of people’s expectations of language models show that their beliefs about model capabilities can diverge from actual performance [54]. Current language models can also fail to update their beliefs about a user’s preferences across interactions [55]. Co-evolution therefore requires tracking changes in both parties and the relationship between them. Task performance, human capability and each party’s understanding of the other should be evaluated as distinct quantities.

3. Third, the allocation of work is a central mechanism through which the agent shapes this trajectory. Self-evolving AI systems divide work into three roles [56]. A proposer decides what the next task is and what would count as success, a solver produces a candidate, and a verifier judges the result and supplies correction. In human–AI assistance, each role can be held by either party and allocated separately at different levels of the same task [57]. At each step, the agent can determine how much of the task to solve, which decisions to return to the person and where to create opportunities for practice. In the tutoring study described above, the same underlying model reduced or largely preserved later unaided performance depending on whether it gave complete answers or hints [10]. In professional work, the same principle may mean that the agent solves implementation details while the person proposes the architecture and verifies the change. These allocations should be revisited as the person’s goals and capabilities change. A novice may first rely on the agent to propose and verify, and later take over these roles as the relevant capabilities develop. The movement of roles between the two parties is therefore part of the trajectory of the coupled system.

These dynamics also change what it means to optimize assistance. A conventional objective can focus on the immediate task outcome, while a developmental objective must also account for the trajectory of the human state. Schematically, this can be written as

$$J = \sum_t (T_t + G_t - D_t),$$

where T_t denotes task outcomes, G_t denotes growth in capabilities the person has chosen to develop as specified by g_t , and D_t denotes unwanted dependence. The relative importance of these terms depends on the person’s goals, and some goals may be better expressed as constraints than as quantities to trade off [58]. A person may also deliberately delegate a capability without seeking growth in it. In that case, reliance on the agent for this capability is not counted as unwanted dependence, as long as the person retains the capability floor described above. The purpose of the formulation is to make the human consequences of assistance explicit as part of the objective of the human–AI system.

This perspective also distinguishes several possible trajectories of the coupled system (Fig. 1d). Sustained co-evolution occurs when the agent adapts to the person while the person’s chosen capabilities also develop. Dependence can emerge when the agent

becomes increasingly effective while the person’s capability to perform or supervise the relevant work stagnates or declines [3, 39]. In terms of roles, dependence appears as a persistent migration of the verifier role to the agent, where the person keeps accepting outputs and stops detecting errors. Deliberate delegation describes a different trajectory, in which the person intentionally relies on a relatively stable capability of the agent while retaining the capabilities needed to set goals, evaluate assistance and supervise the delegation. The relationship can also stall when neither side adapts. These regimes are a diagnostic vocabulary rather than a finished taxonomy. Longitudinal evidence is needed to determine whether they correspond to meaningful outcomes and to distinguish dependence from deliberate delegation or temporary stages of learning. The central divergence is that assisted performance can increase while unaided human capability remains flat or declines (Fig. 1c).

5 Challenges for developmental alignment

Developmental alignment requires solving three problems that current AI systems are not designed to address together. We need to measure how assistance changes a person’s capabilities, determine who has the authority to define which changes are desirable, and learn how different forms of assistance affect that person over time.

Measurement

The first problem is separating what the person can do from what the human–AI system can accomplish together. As noted earlier, a person may complete a difficult task with an agent while gaining little ability to perform the work independently. Conversely, an agent may make a task easier without reducing the person’s underlying capability. Assisted performance therefore provides an incomplete measure of human capability.

The most direct evidence comes from what happens after assistance is removed. Delayed, unaided performance on tasks that require transfer can reveal whether a capability has changed beyond the assisted interaction [7, 10, 59]. The relevant tests depend on the capability being developed. A developer who wants to review generated code should be able to detect errors and modify a solution without the agent. A student who wants to understand a concept should be able to retrieve and apply it later. In other domains, the corresponding test might involve independent judgement, decision-making or supervision.

Measurement also creates a potential failure mode for the system itself. Once a capability measure becomes part of an optimization target, an agent may learn to improve the measure without producing the capability it is intended to represent, as described by Goodhart’s law [60] and illustrated by research on reward hacking and misspecification [61, 62]. Evaluation therefore needs some independence from the agent’s optimization. At the same time, repeated measurement imposes costs on the person. Frequent testing consumes attention, can itself change the capability being measured [45] and raises privacy concerns already identified for personalized language models [5]. The practical challenge is to obtain credible evidence of human change across time and contexts without turning assistance into continuous assessment.

Authority

Measuring change does not tell us whether the change is desirable. A capability can increase while becoming less relevant to what a person values, and a capability can decline because the person has deliberately chosen to delegate it. Developmental alignment therefore needs a source of authority for deciding which capabilities should be retained, developed or delegated. That authority ultimately belongs to the person, subject to legitimate constraints in contexts where others have a recognized role.

This creates a further problem for AI systems. The goals that determine whether assistance is successful are themselves exposed to the system’s influence. An agent that can persuade a person to change those goals can make its own objective easier to satisfy. In terms of roles, the agent would then take over the proposer role at the level of the person’s goals. This resembles reward tampering, where an agent changes the mechanism used to evaluate its behaviour [63], and relates to work on aligning AI with reward functions that can themselves be influenced [14]. Related work shows that recommenders can shift user preferences [16] and that models trained on user feedback can learn to target such changes [17]. Developmental goals therefore need to be explicit, revisable by the person and protected from being silently rewritten through the assistance process.

The problem becomes harder when authority is shared. A teacher may set learning requirements for a student, a clinician may recommend goals relevant to treatment, or an employer may define capabilities required for a job. These constraints can coexist with the person’s own developmental goals, and commercial systems introduce additional incentives around engagement and approval [64]. If those incentives become the effective objective, an agent may favour assistance that increases reliance or approval even when it conflicts with the person’s chosen development [18, 37, 65]. Developmental alignment therefore needs explicit mechanisms for representing authority, handling disagreement and allowing the person to inspect and revise the goals that govern assistance.

Learning

Once the goals and relevant capabilities are known, the agent still has to decide how much help to provide. Solving a problem, giving a hint, asking the person to attempt it first or returning a verification step can produce different effects on the same task. The agent therefore needs to learn action-conditioned predictions about both task outcomes and human outcomes. It does not need a complete psychological model of the person. It needs enough information to estimate how alternative forms of assistance are likely to affect capabilities that person has chosen to develop.

The main difficulty is that these effects may be delayed. A choice made today may affect what the person can do weeks later, and the intervening interactions will have changed the human state from which those effects are observed. Standard action-outcome learning therefore faces a moving target. World-model research provides methods for predicting the consequences of actions [66, 67], while learner models provide tools for estimating changing capability [52]. Long-horizon credit-assignment methods address delayed effects [68], but none of these lines of work by itself establishes how a particular person’s capability changes under different assistance policies.

Learning this relationship from real users also has a cost. Simulated users can make it possible to compare assistance policies at low cost, but they are imperfect proxies for human adaptation [69]. In a randomized medical study, neither benchmarks nor simulated patients predicted how people actually fared with language-model assistance [70]. Real-user experimentation can provide the missing evidence, but it consumes people’s time and trust and can expose them to poorly chosen forms of assistance. Systems that learn how to assist therefore need bounded exploration, consent and safeguards against harmful experimentation [71].

These three problems form a single loop. We need measurement to observe how the person changes, authority to determine which changes matter, and learning to connect assistance decisions to those changes. Failure at any point can undermine the others. Developmental alignment therefore calls for a research programme that connects these three components while leaving the particular agent architecture open.

6 From human development to AI development

The consequences of developmental alignment may extend beyond the individual relationship. As AI systems increasingly participate in creating and improving future AI [72], the capabilities people develop through AI assistance may also shape how those systems are built, evaluated and governed. This creates a broader cycle:

$$\text{AI} \longrightarrow \text{Human} \longrightarrow \text{AI}.$$

We call this possibility *AI for Human for AI*. The idea is not that human development should serve the development of AI. Rather, AI systems still depend on human expertise, judgement and direction, including the supervision of models that may become more capable than their supervisors [73]. Changes in human capability can therefore affect the future systems that people create and oversee.

This possibility is becoming concrete as AI enters scientific and technical work. Automated research systems can already generate ideas, run experiments and produce research outputs [72], while AI collaborators are being developed to work iteratively with human researchers [74]. AI adoption can increase individual research productivity and citation impact, while also narrowing the range of topics explored across the research community [75]. At the same time, researchers have warned that AI assistance can create an illusion of understanding [76]. These developments make the capabilities retained by people relevant to the future of AI itself. If people increasingly delegate scientific judgement, technical understanding or evaluation to AI systems, the human capacity to direct and scrutinize future AI may also change [77].

This broader implication reinforces the central argument of this perspective. AI assistance changes the human state through the work it performs, and those human changes can affect future interactions, institutions and AI development. Developmental alignment therefore concerns more than preserving performance in individual tasks. It concerns whether people retain the authority and capabilities they choose to cultivate as AI becomes embedded in their activities. Human–AI co-evolution describes the resulting trajectory, while developmental alignment provides a way to shape that trajectory according to human-defined goals.

Declarations

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